Shifting Gears, Smashing Carbon: Towards Net-Zero with Tidal Wave



OPS-COGITATE

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TEAM STRATEGIZERS

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EXECUTIVE SUMMARY

Situation

1. **Key Priority**: Develop a robust baseline of current emissions across all scopes (1, 2, and 3) within the supply chain
2. **Project Sustainify:** Initiation to quantify and analyze emissions from Tidal Wave's operations, suppliers, and customers

TIDAL WAVE

Complication

LIMITED MANAGERIAL AWARENESS

Hindering the development of an effective strategy for decarbonization

REGULATORY LANDSCAPE

Difficulty in maintaining efficient emissions control and creating a lack of transparency

Question

"How can the company achieve its targets to **decarbonize** by contacting to its stakeholders to obtain comprehensive data enabling the integration of a low-emission strategy under **Project SUSTAINIFY?**"

Answer

TAILORED ENGAGEMENT

Decarbonization hinges on tailored communication, utilizing platforms, and digital solutions for consumer engagement

PRODUCT INNOVATION

Invest in low-carbon braking systems, in recycled materials, offer "green" product lines which incentivize emissions reduction

GREEN EXPANSION

Prioritize renewable builds, sustainable markets & tech acquisitions to accelerate eco-braking leadership

STRATEGIC EXPANSION PLAN

- **FY-0** — KNOW YOUR CARBON
- **FY-1** — MAP AND ENGAGE SUPPLIERS
- **FY-2** — STREAMLINE DATA
- **FY-3** — PARTNER TO REDUCE
- **FY-4** — TRANSPARENCY AND IMPROVEMENT

SCOPE 3 REPORTING (2023)

20% — Increase in revenue of companies with high end execution

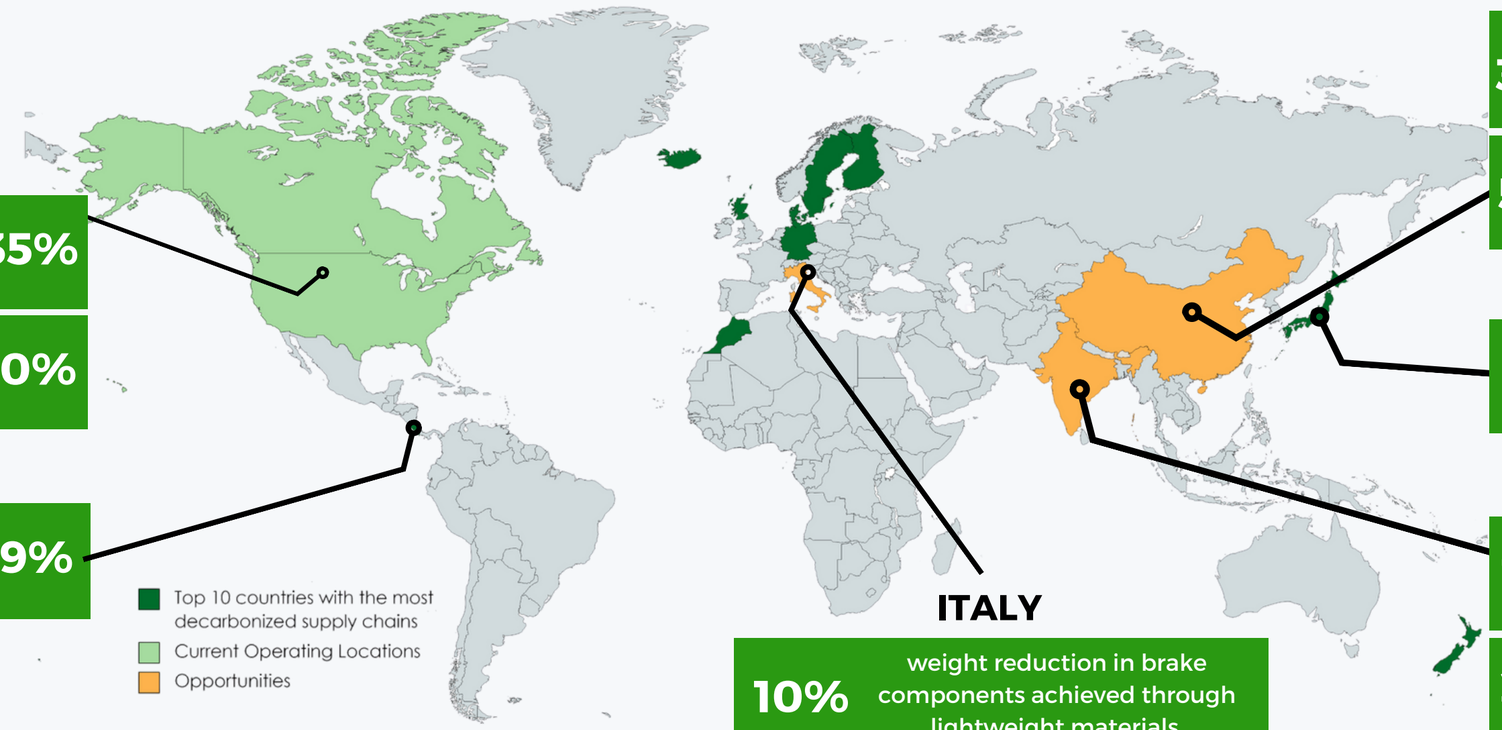
38% — of businesses track their supply chain emissions

Revenue Projections (FY 2023-30)

BACKGROUND

- Company is based in US and Canada
- Scope-3 Reporting is not carried out
- Rigidity regarding Conventional Measures
- No sustainability measures implemented

TRENDS & COMPLEXITIES



USA

35% of new car models featuring ADAS with regenerative braking capabilities

20% of brake manufacturers initiating closed-loop recycling programs for brake pads and discs

COSTA RICA

19% reduction in GHG emissions targets set up by NDC under the Paris Agreement by 2030

ITALY

10% weight reduction in brake components achieved through lightweight materials.

CHINA

30% increase in EV production capacity since 2020, driving demand for regenerative braking systems

50% of brake pad manufacturers are implementing low-dust and low-noise materials by 2025

JAPAN

5% of new vehicles expected to be H fuel cell powered by 2030, requiring specialized braking systems

INDIA

10% reduction in carbon footprint targeted through green logistics initiatives in the supply chain

25% growth in the auto parts market by 2025, with increased focus on sustainable materials & practices

PITFALLS OF SCOPES ASSESSMENT

DIRECT

- Operational Emissions from **Manufacturing Facilities**
- Emissions from **Vehicle Fleet** for transportation and delivery

INDIRECT

- **Energy-intensive** nature of manufacturing braking systems
- Carbon Intensity of **Purchased Electricity**

OTHER INDIRECT

- Globally **dispersed** Supply Chain
- Collaboration with **OEMs**
- **Disposal and recycling** at the end of their life cycle
- Suppliers Engagement

PATHWAY TO GREEN PLAN

ACTION PLAN

Transparency & Improvement

FY-4

- Standardized reporting for **internal and external stakeholders**
- Independent verification of our emissions data
- Sustainability report showcasing progress and **future goals**

FY-3

Partner to reduce

- Model different decarbonization pathways
- Collaborate with **key suppliers on green initiatives**
- Industry partnerships for wider impact

FY-2

Streamline Data

- **Pilot test data collection tools** with select suppliers
- Implement standardized system across the chain
- Partner with tech providers for automation



FY-1

Map & engage suppliers

- Support high-impact suppliers with data collection
- **Tiered engagement** strategy for diverse suppliers

FY-0

IMMEDIATE MEASURES

Know Your Carbon

- Train everyone on emissions (Scopes 1, 2, 3)
- Build a **decarbonization team**
- Internal carbon pricing to incentivize reductions

SCOPES PLAN

SCOPE 1 — Direct Emissions



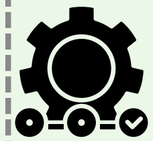
- Map direct emission sources (boilers, vehicles)
- Implement monitoring systems and data collection processes
- Identify and implement emission reduction strategies

SCOPE 2 — Indirect Purchased Electricity



- Obtain supplier data on electricity generation sources
- Explore renewable energy options
- Advocate for cleaner electricity generation through policy engagement

SCOPE 3 — Other Indirect Emissions



- Identify key suppliers & customers contributing to emissions
- Develop incentives and support mechanisms for data sharing
- Define consistent data collection formats and reporting protocols
- Assess supplier and customer emission profiles and identify hotspots
- Work with partners to implement joint decarbonization initiatives

PESTEL ANALYSIS

P

- **Climate Policies:** Incentives for decarbonization, carbon pricing
- **Govt Support:** Funding for green technologies, infra development

E

- **Market Volatility:** Fluctuations in resources & energy prices impacting production
- **Investor Pressure:** Rising demand for sustainable investments

S

- **Labor Rights & Standards:** Ethical sourcing, human rights concerns in supply chains
- **Community Engagement:** Local stakeholder input, potential conflicts over resource use

T

- **Advancements in GreenTech:** Efficiency improvements, carbon capture & storage sols
- **Data Acquisition & Mngmt:** Tools for emissions tracking, transparent reporting platforms

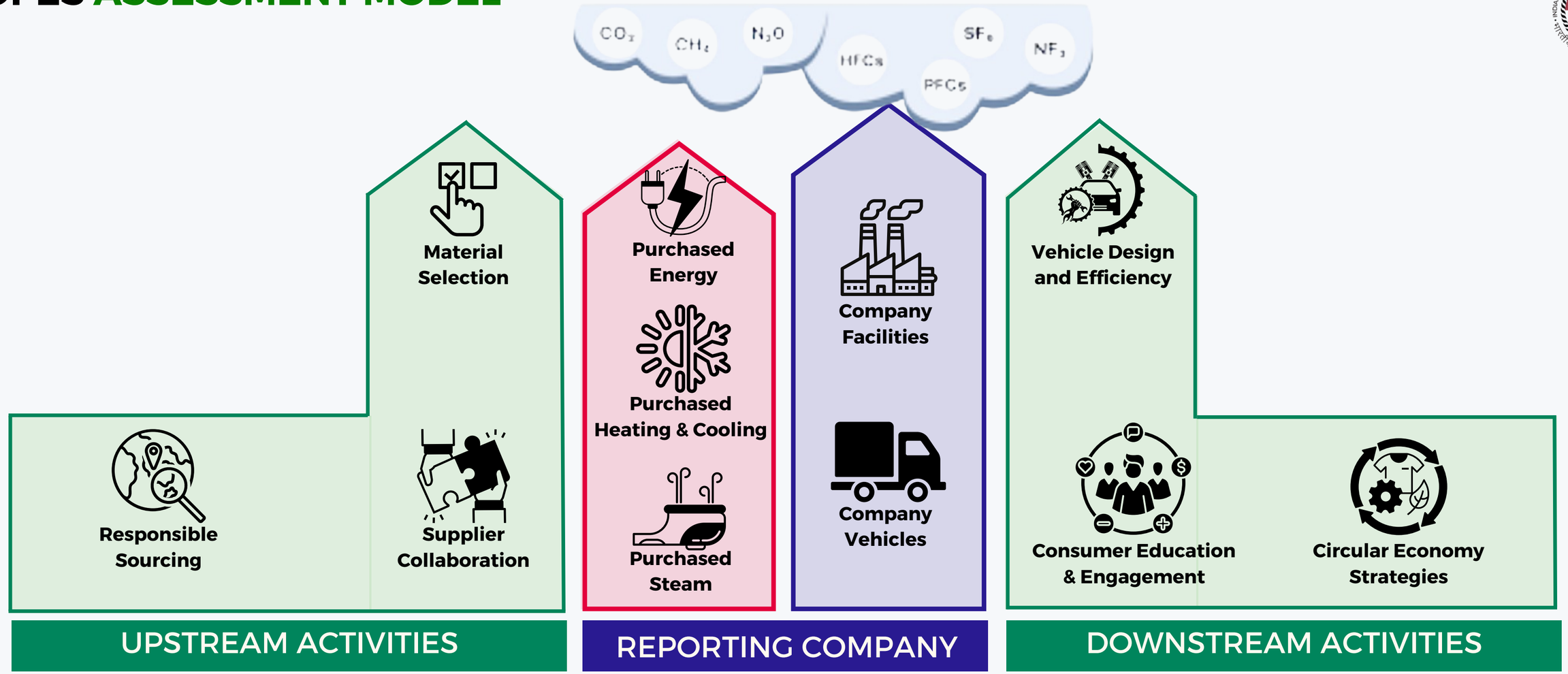
E

- **Climate Change Impacts:** Resource scarcity, extreme weather events
- **Regulatory Pressure:** Environmental compliance, CFP reduction targets

L

- **Product Liability:** Potential lawsuits related to environmental impact
- **Greenwashing Regulations:** Increased scrutiny of sustainability claims, reporting standards

SCOPES ASSESSMENT MODEL



• Sustainable supply chain mapping • Ethical sourcing practices	RESPONSIBLE SOURCING	VEHICLE DESIGN AND EFFICIENCY	• Lightweighting and aerodynamic optimization • Regenerative braking technology adoption	• Advanced Driver Assistance Systems (ADAS) integration
• Collaborative emission reduction initiatives • Investment in low-carbon technologies • Transparency and data sharing • Supplier certification and traceability	SUPPLIER COLLABORATIONS	CONSUMER ENGAGEMENT	• Eco-driving education and awareness campaigns • Consumer product life cycle transparency	• End-of-life options promotion and information
• Low-embodied carbon material selection • Alternative braking system exploration • Design for material efficiency	MATERIAL SELCTION	CIRCULAR ECONOMY STRATEGIES	• Design for disassembly and recycling • Closed-loop material recovery infrastructure	• Extended producer responsibility initiatives

OUTCOME JOURNEY

BRAND & REPUTATION — Enhance brand image as **a sustainability leader**, attracting environmentally conscious customers & investors

FINANCIAL GAIN — **Access green financing** and potential cost advantages from carbon pricing schemes

REGULATORY COMPLIANCE & RISK MITIGATION — Reduce risk of **non-compliance penalties** and reputational damage

SUPPLY CHAIN RESILIENCE — Enhance collaboration and partnership with suppliers based on **shared sustainability goals**

SOCIETAL & ENVIRONMENTAL CONTRIBUTION — Promote sustainable practices and **innovation** in the transportation sector

FASTER CLIMATE ACTION — Comprehensive data informs **targeted strategies** for faster decarbonization across the supply chain

ANNUAL COST SAVINGS — Increased energy efficiency and optimized processes lead to **tangible financial benefits**

GREEN FINANCING ACCESS — Quantified emissions reductions **unlock greener loan options** and lower borrowing costs

EMPLOYEE PRIDE — Active involvement in sustainability initiatives **fosters employee engagement** and satisfaction

SUSTAINABLE FUTURE — Leadership in reduced emissions contributes to a cleaner, **healthier planet for all**

ANALYZING GROWTH DRIVERS

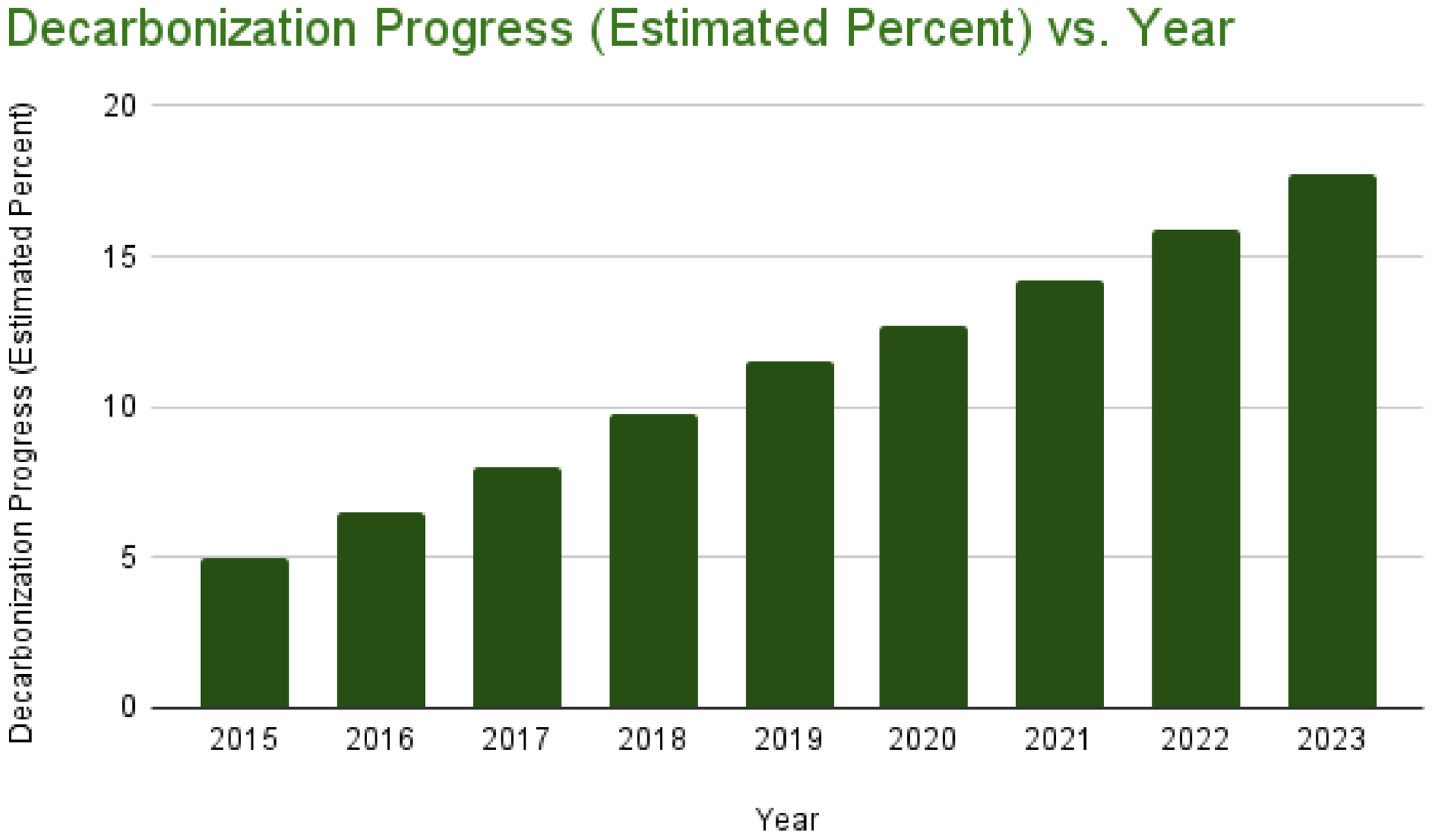
GROWTH TRIGGERS	IMPACT
Successful **Scope 3 Reporting** & **Supply Chain Decarbonization**	
Innovation in **Sustainable** Braking Technology	
Strategic Expansion into Emerging Markets	
Fostering **Strong Customer Relationships** and Brand Loyalty	
Effective **Digital Marketing** and Branding Strategies	
Developing Aftermarket Solution	


THANK YOU



APPENDIX

APPENDIX


Decarbonization Progress (Estimated Percent) vs. Year
Decarbonization Progress (Estimated Percent)
20
15
10
5
0
2015
2016
2017
2018
2019
2020
2021
2022
2023
Year

APPENDIX

Country	Target Year
India	2070
China	2060
European Union	2050
United States	2050
United Kingdom	2050
Canada	2050
Japan	2050
South Korea	2050
Brazil	2050
Argentina	2050
Chile	2050
Costa Rica	2050
New Zealand	2050

SOURCES & REPORTS

- Automative:Pathways to Decarbonization: Deloitte
- Reducing carbon footprint in the automotive industry: Grant Thornton
- Green Hydrogen:Energizing the path to net zero: Deloitte
- Decarbonizing growth: KPMG
- The future of Supply Chains: KPMG
- Next Generation Supply Chains: Supply Chains 2020: Mckinsey
- Building the Supply Chain: BCG
- Tech Trends 2024: Deloitte
- EY Sustainability Report 2022: EY
- Digital trends in supply chain survey: PwC